

ENGR 3323: Signals and Systems

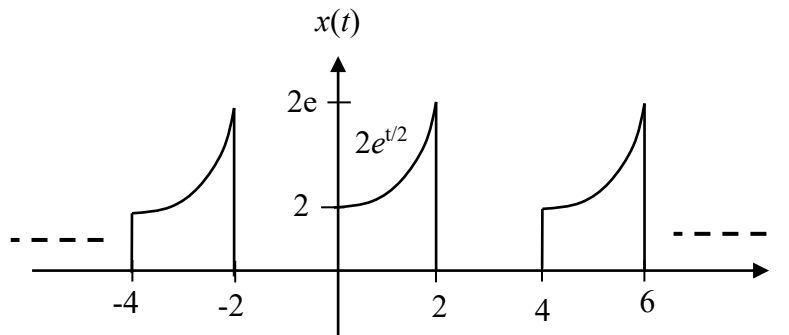
Sample of the Final (Test 4)

Time:

Name:

Please submit your solution in D2L and the hard copy to the teacher.

- 1) For the periodic signal $x(t)$ shown below
- Find the exponential Fourier series coefficient D_n .
 - Find the powers of the DC and the second harmonica.



- 2) The signal $x(t) = \sum_{n=1}^{\infty} \frac{2}{\pi n^2} \cos(n2t)$ is applied as an input to a system with transfer function
- $$H(\omega) = \frac{2}{4 + j2\omega}$$

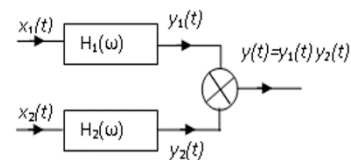
Find the output $y(t)$. Make sure your final answer does not have j in $y(t)$.

- 3) Find the Fourier Transform $X(j\omega)$ of the signal $x(t)$

$$x(t) = 4e^{-2t} \quad \text{for } 0 \leq t \leq 2$$

- 4) Signals $x_1(t) = 100 \text{ rect}(100t)$ and $x_2(t) = \delta(t)$ are applied at the inputs of the ideal lowpass filters $H_1(\omega) = \text{rect}(\omega/800\pi)$ and $H_2(\omega) = \Delta(\omega/400\pi)$, see below. The outputs $y_1(t)$ and $y_2(t)$ of these filters are multiplied to obtain the signal $y(t) = y_1(t)y_2(t)$.

- Find and Sketch $X_1(\omega)$ and $X_2(\omega)$
- Sketch $H_1(\omega)$ and $H_2(\omega)$
- Sketch $Y_1(\omega)$ and $Y_2(\omega)$
- Find the bandwidth of $y_1(t)$, $y_2(t)$, and $y(t)$



- 5) A TV signal has a bandwidth of 4 MHz. This signal is sampled, quantized, and binary coded.
- Determine the sampling rate if the signal is to be sampled at a rate 20% above the Nyquist rate.
 - If the samples are quantized into 512 levels, what number of binary pulses is required to encode each sample?
 - Determine the binary pulse rate (bits/s) of the binary-coded signal.
 - If the amplitude range of the TV signal is $[-2, 2]$ volt, what is the maximum quantization error?